The dog game: frequently asked questions

Charith Reddy Pareddy · November 3, 2026 · figures refer to Dog_Game_Results_and_Updates.pdf

The game

What is the dog game?

The Battling Influencers Game (BIG; Wu, Zhu, Cai and Zhu, 2025) in the unit square. Influencers with targets report points, a receiver (the dog) moves, and each influencer loses the squared distance between the dog and its target, discounted over rounds. In BIG the dog goes to the weighted mean of the reports.

What is added to BIG here?

Inertia. The dog keeps part of its old position: $x_{t+1} = \lambda\, x_t + (1 - \lambda) \times$ aggregate. At $\lambda = 0$ this is BIG; near 1 the dog is very slow. It describes things that cannot jump, such as an opinion, a price or a robot with a speed limit.

Why is this a Markov game?

Today's reports change where the dog is tomorrow, so influencers must think ahead. The dog's position is the state, and it is all an influencer needs to choose its report.

Why do influencers exaggerate?

A report counts for only a share of the average, so an influencer overshoots to pull the average toward its target. In BIG's equilibrium all influencers but at most one report at the edge of the box.

Why solve one coordinate at a time?

Losses add across coordinates and the dog moves coordinate by coordinate, so the square splits into two one-dimensional games. That is why x and y can behave differently. The medians here are coordinate-wise, not a rotation-invariant two-dimensional median.

Definitions

What is the one-shot outcome d*?

Where the dog sits in BIG's equilibrium without inertia.

What is a pivot?

The one influencer, if there is one, whose one-shot report is strictly inside the box. Its report puts the dog exactly on its own target, so d* equals that target.

What is a phantom?

A fixed point added to the reports before taking the median. No influencer controls it.

What is a generalized median?

The median of the reports and N - 1 phantoms. With equal weights, averaging's outcome matches phantoms at 1/N, ..., (N - 1)/N, and the plain median has its phantoms at the edges. So one family covers both receivers.

What is the median of two numbers?

It is not unique: any point between them is a median. Taking the midpoint gives the mean. A strategy-proof version adds one phantom α and uses med(a, b, α), a median of three numbers.

So with two influencers, are the median and averaging the same?

With the midpoint rule and equal weights, yes: it is the same game. The outcome is still not the mean of the targets. For targets 0.2 and 0.9 it is 0.5, the median of 0.2, 0.9 and 1/2. That is why the comparisons use three or more influencers.

What is a feedback (Markov perfect) equilibrium?

Reports depend only on the dog's position, and at no position can one influencer gain by changing its report, given how play continues.

What is target form?

Each influencer aims the next dog position at a fixed point, as far as it can reach. That point can differ from its own target, because it plans ahead.

What is a rest point?

The position where the dog stops moving.

The plots

What do the vector fields show?

Fig. 1 (left) shows where the dog moves next from each position. Fig. 2 shows, at each dog position, the direction of each influencer's equilibrium report. Arrow lengths are scaled per panel in Fig. 1 and fixed in Fig. 2.

Why does the averaging path bend?

The coordinates finish at different times. In Fig. 1 y lands on 0.55 in round 5, while x is still moving toward 1/3, so the path comes in diagonally and then runs sideways.

How do averaging and median reports differ?

In Fig. 2 averaging keeps reports near the edges, with 2 or 3 distinct reports per influencer across the square. Median reports move with the dog (8, 5 and 5 distinct reports).

Do the plots prove anything?

No. They show one game with three influencers at inertia 0.8. The general statements are the propositions.

The main new result

What is the new result?

Proposition 19: a sticky receiver reaches its rest point exactly, in finitely many rounds, when an influencer's target decides that point. When a phantom decides it, the receiver only gets closer and never arrives.

Why does this happen?

A phantom is fixed, so inertia shrinks the gap by a factor λ every round. An influencer recomputes its report every round to make up for inertia, so a pivotal influencer cancels it and lands the dog on its target.

Does never arriving matter if the gap is tiny?

Often the gap is small: in Fig. 3 it is 0.0023 after 25 rounds. The point is the qualitative split, which identifies what decides the end of the path and gives exact arrival counts. It does not by itself say one receiver has lower loss.

Which receivers arrive exactly?

Averaging arrives only with a pivot. The median arrives unless the median target is 0 or 1. Compensated receivers arrive at any interior rest point, even a phantom, because they clip the whole aggregate to what they can reach.

How many rounds does it take?

Proposition 18 gives $\text{ceil}(\log((U - x)/(U - d^*)) / \log(1/\lambda))$ from $x < d^*$, with U the dog when the pivot reports 1, and a mirror formula above d^*. With equal weights, targets 0.1, 0.5, 0.9 and inertia 0.8, that is 7 rounds from $x = 0$.

How common is a pivot?

It appeared in 115, 96 and 100 of 200 random games for 3, 5 and 7 influencers (numerical).

Is the policy in Proposition 18 always an equilibrium?

No. It is proved when every edge influencer is far enough from the pivot, and that condition is rarely met at high inertia. The condition is sufficient, not necessary. When it fails, edge influencers usually gain by easing off, by up to 0.077.

Does inertia change where the dog ends?

In Propositions 1, 2 and 18 it does not: the dog ends at d^* at every inertia, and long-run losses in the saved games do not change with inertia. Inertia changes how fast it gets there.

Does this hold in continuous time?

As a limit: with time step $1 - \lambda$, the arrival time tends to $\ln((U - x)/(U - d^*))$, which is finite, while a phantom gives only an exponential approach. That describes the limit of the discrete game; an equilibrium of the continuous-time game is not proved.

The averaging equilibrium

When can influencers just keep their one-shot reports?

Proposition 1: exactly when no one-shot action is inside the box and every edge influencer's slack $(t - d^*)/(1 - d^*)$ is at least $\rho(\lambda, g) = \lambda(1 - g\lambda)/(1 - g\lambda^2)$. The threshold rises with inertia.

What does the two-player equilibrium look like?

Proposition 2: when the targets straddle d^*, each player aims the next position at $d^* + k(t - d^*)$, with $k = (1 - g\lambda^2)/(1 - g\lambda)$. It held in 77 of 77 random straddling games.

Do players aim past their own targets?

Yes, because k is at least 1. The factor is largest at an intermediate inertia, where it is at most 2.

Other receivers

Why are median-type receivers not the focus?

Trimmed mean and median-type mechanisms have been studied before, so here they appear as comparisons. The focus is averaging in the dynamic game.

Is the median better than averaging?

It depends on inertia. Without inertia it wins in about a fifth of random games; at inertia 0.9 it wins in 81 to 88%, for 3, 5 and 7 influencers (numerical, 100 games per point). The cause is speed, not location.

Is the median still strategy-proof with inertia?

No (proved). It moves only part of the way, so the pivotal influencer over-reports to make up for it.

What about the trimmed mean?

A small check with five influencers found many equilibria, because a trimmed influencer has no influence. In the equilibrium with reports at the edges, no influencer gained by deviating in the games checked.

What is a compensated receiver?

A receiver that moves to the reachable point closest to a strategy-proof rule's outcome. It keeps truthful reporting an equilibrium if it can stay put and its reach does not fall (Theorem 1, Propositions 15 and 16). This is a side thread.

What does learning find?

Self-play finds the one-shot equilibrium (74 of 75 REINFORCE runs, 25 of 25 PPO runs). With a persistent dog, A2C and PPO miss the feedback equilibrium in 19 of 20 runs. Fitting each state's exact best response cuts the largest gain to at most 0.001 (5 of 5 seeds).

Evidence and limits

What is proved and what is numerical?

Proved: Propositions 1, 2, 18 and 19, and the median's loss of strategy-proofness. Numerical: the median-against-averaging comparison, pivot frequencies, arrival when the condition of Proposition 18 fails, and all learning results.

How was Proposition 19 checked?

It has a written proof and a test, and it held in 3,000 random cases.

What was corrected?

Theorem 3 was narrowed: the best compensated phantoms do not depend on inertia when each phantom's criterion changes sign once, as for uniform targets. The shading gain under clipped shocks is about 0.0015, not 0.0024, and about 0.0001, not 0.0003; the old values were quadrature error.

What is still open?

The general pivot case: target form was an equilibrium in 10 of 12 checked games, and in the other 2 an influencer gains up to 0.005 because its objective has several peaks. Also open are uniqueness of the feedback equilibrium, unequal weights in Proposition 19, and a rest point that is both a target and a phantom.

Are these results new?

As far as the searches went, no earlier work combines strategic reports, a persistent median-type receiver and the split between targets and phantoms. Mei et al. (2022) have finite-time convergence and Cheng and Mei (2026) an inertial median, but their agents are not strategic. Wada et al. (2018) prove a special case of Theorem 1. The searches were not a systematic review, the mathematics of Proposition 19 is elementary, and none of the proofs has been refereed.

Where are the proofs and tests?

The proofs are in docs/INERTIA_PROOFS.md and the tests in tests/test_mechanisms.py.